

Mitochondria Present in Media of Cultured Human Skeletal Muscle Cells

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INTRODUCTION

Despite traditionally being viewed as solely intracellular units, mitochondria are now understood to also exist outside the cell, as extracellular mitochondria. Extracellular mitochondria have been reported to exist as either Free mitochondria, or as mitochondria encapsulated within an extracellular vesicle. Extracellular mitochondria have been extensively documented as the product of active secretion of mitochondria in a range of different types of cells, and secretion of mitochondria is thought to be dependent on the reorganisation of cytoskeletal components such as actin and tubulin filaments, and promoted by elevated intracellular calcium. However, despite the high mitochondrial density of skeletal muscle, and the central role of calcium in muscle contraction, mitochondrial secretion remains poorly studied in skeletal muscle cells. Therefore, the aim of this study was to investigate whether human skeletal muscle cells are capable of secreting mitochondria.

METHODS

Human primary muscle cells were cultured and differentiated to form myotubes, and subjected to a 24-hour incubation in the presence of 5 mM caffeine to stimulate intracellular calcium release, and/or inhibitors of the cytoskeletal components actin and tubulin. Mitochondria were isolated from cell culture media with magnetic immunoprecipitation to select for the mitochondrial outer membrane protein TOMM22, and quantified with the MitoTracker Green fluorescence assay. Western blot was performed to characterise the presence of inner-membrane (mitofilin) and matrix (citrate synthase) mitochondrial proteins, as well as the presence of the extracellular vesicle marker CD9.

RESULTS AND DISCUSSION

Mitochondria were successfully isolated from cell culture media. Notably, mitochondrial content was reduced in media from cells treated with 5mM caffeine, contrasting findings from studies in other cell types. Additionally, none of the treatments significantly increased lactate dehydrogenase in media, which is a marker of cell damage. This supports the phenomenon of regulated secretion of mitochondria, as opposed to uncontrolled leakage from the cell due to cytotoxic effects and a compromised plasma membrane.

Preliminary results from western blotting found presence of the mitochondrial matrix protein citrate synthase in mitochondrial pellets, supporting the possibility that intact mitochondria are secreted from skeletal muscle cells. As this study remains ongoing, results are not yet collected in full.

CONCLUSIONS

This study demonstrated that mitochondria are present in the media of cultured human skeletal muscle cells. The presence of mitochondria in culture media was reduced in cells treated with caffeine, and there was no evidence of muscle damage, supporting a regulated and mechanistic process of mitochondrial secretion.

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